

Abeer Sethia

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AI Research Engineer with hands-on experience in medical image segmentation, self-supervised learning, and clinical pipeline validation. Built a **real-time liver segmentation system** at **Philips** validated on 500+ clinical scans, developing a **trackerless fetal ultrasound pose estimation pipeline** at **IISc Bangalore**, and engineered **physics-informed deep learning models** for **signal reconstruction** at **IIT Kharagpur**.

EDUCATION

Manipal Institute of Technology, Manipal

2022–2026

Bachelors of Technology - Data Science and Engineering, Minor in Multimodal Intelligent Systems

Courses: Computer Vision, Natural Language Processing, Deep Learning, Statistics, Mathematical Foundations

SKILLS

- **Deep Learning:** CNNs, RNNs, LSTMs, GNNs, Attention Mechanisms & Transformers, U-Nets, VLMs
- **Computer Vision:** Classification, Segmentation, Object Detection, Camera Geometry
- **Natural Language Processing:** Named Entity Recognition (NER), Multi-Label Sentiment Analysis, NLI Ensembling
- **Frameworks & Tools:** PyTorch, PyTorch3D, PyG, TensorFlow, Keras, CUDA, OpenCV, ITK, MONAI, Git, Docker
- **Languages:** Python, MATLAB, C++, Java, R, Bash (Linux), SQL, $\text{\LaTeX}$
- **Distributed Systems:** Hadoop, Apache Spark, Hive, Pig

RESEARCH EXPERIENCE

Research Intern – AI, UTSAAH Lab, Indian Institute of Science Bangalore 🔗

December 2025–Present

- Developing a trackerless fetal ultrasound pipeline using **contrastive self-supervised learning** to extract robust temporal representations from B-mode sequences.
- Engineering an $SE(3)$ **pose estimation network** to predict pose from short frame windows, validating trajectories against **optical and EMT tracker ground truth**.

Research Intern - AI, KLIV Lab, Indian Institute of Technology Kharagpur 🔗

April–October 2025

- Engineered a temporal CNN autoencoder to **reconstruct 3D chaotic attractor dynamics from a single-channel** observable, achieving **98.3%** (X/Y) and **93.4%** (Z) phase-space correlation at an 8:1 compression ratio.
- Formulated a **differentiable False Nearest Neighbors (FNN) loss** function to enforce causal consistency and global invariants, preventing trajectory self-intersection during multi-dimensional embedding trajectory recovery.

PROFESSIONAL EXPERIENCE

Technical Intern, Philips 🔗

March–July 2025

- Engineered a real-time **Flask inference pipeline** for liver lesion segmentation, deploying fine-tuned **nnU-Net** and **MedSAM2** models validated on **500+ clinical CT/MRI scans**.
- Achieved **0.92 Dice (0.89 IoU)**, outperforming standard U-Net baselines by **12%**.

PROJECTS

HVQ-AP: Hippocampal Volume Quantification for Alzheimer's Progression [GitHub](#) 🔗

- Quantified hippocampal volumetric progression in Alzheimer's patients from 3D MRI by building an end-to-end pipeline covering DICOM ingestion (PyDICOM), 3D CNN segmentation, and longitudinal volume tracking.
- Achieved **Dice 91.6% and Jaccard 84.6% on the Medical Segmentation Decathlon benchmark**, matching clinical-grade validation standards.

DropEdge++ with Johnson–Lindenstrauss Projection (Minor Specialization Project) 🔗

- Developed a GNN framework combining **topology-aware edge dropout** (DropEdge++) with mathematical-based random feature projection to mitigate oversmoothing and improve computational efficiency.
- Achieved up to **8% accuracy gains** and **3× faster training on the Cora citation network benchmark**.

RAG Based Chatbot for Research Papers [GitHub](#) 🔗

- Built an end-to-end RAG ML pipeline (ingest → chunk → embed → index → retrieve → generate) indexing research PDFs into **65 Pinecone vectors/paper with 384-dim embeddings and Top-5 retrieval**.
- Hardened ML ops for reliability: batched inference/upserts (32/100), local embedding serving migration, and vector validation guards preventing bad writes; served responses via Streamlit/CLI with configurable inference parameters.

CO/EXTRA CURRICULAR

- **Expertise Head – Research Society Manipal:** Mentored 20+ students in research and scientific writing.
- **Data Analyst – IAESTE LC Manipal:** Built automation workflows and dashboards for **1,000+ members**, reducing manual effort by **70%** and improving outreach efficiency by **30%**.